

Personal data & AI

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About me



Shall we play a
game?



About the lecture

- Personal data and how is it regulated in the UK
- ICO's role and actions in the AI space
- Explore AI from a data protection lens

- Q&A time
- Group activities

Quiz time



What is personal data and how is it regulated?

‘personal data’ means any information relating to an identified or identifiable natural person (‘data subject’); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person

Article 4 (1) UK GDPR

Controllership

‘controller’ means the natural or legal person, public authority, agency or other body which, alone or jointly with others, determines the purposes and means of the processing of personal data

Article 4 (7) UK GDPR

Data protection principles



Fairness, lawfulness and transparency: Personal data must be processed fairly, lawfully and transparently



Purpose limitation: Personal data must be processed for specific, explicit and legitimate purposes



Data minimisation: Personal data must be adequate, relevant, and limited to what is necessary



Accuracy: Personal data must be accurate and, where necessary, kept up to date



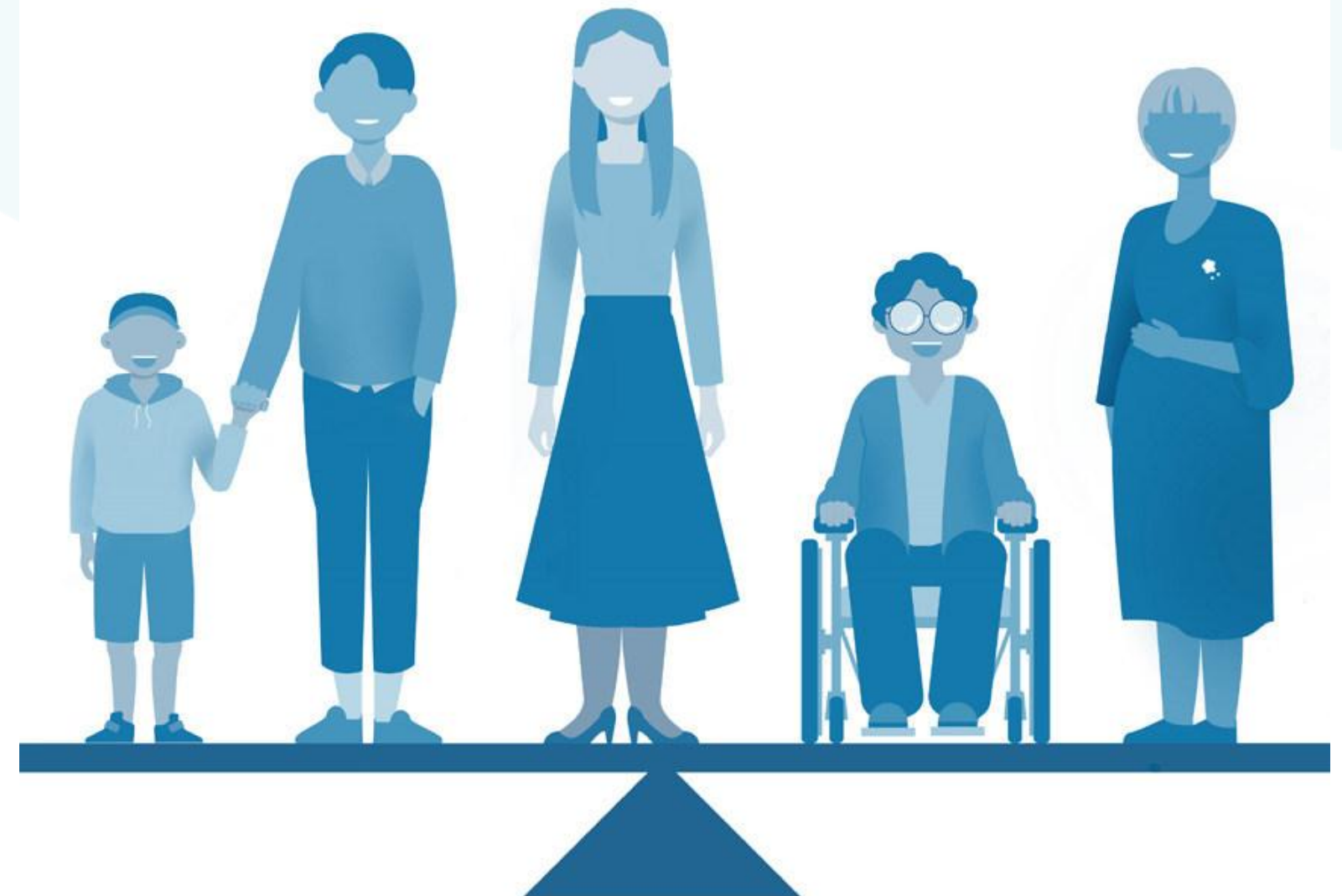
Storage limitation: Personal data must not be kept for longer than is necessary



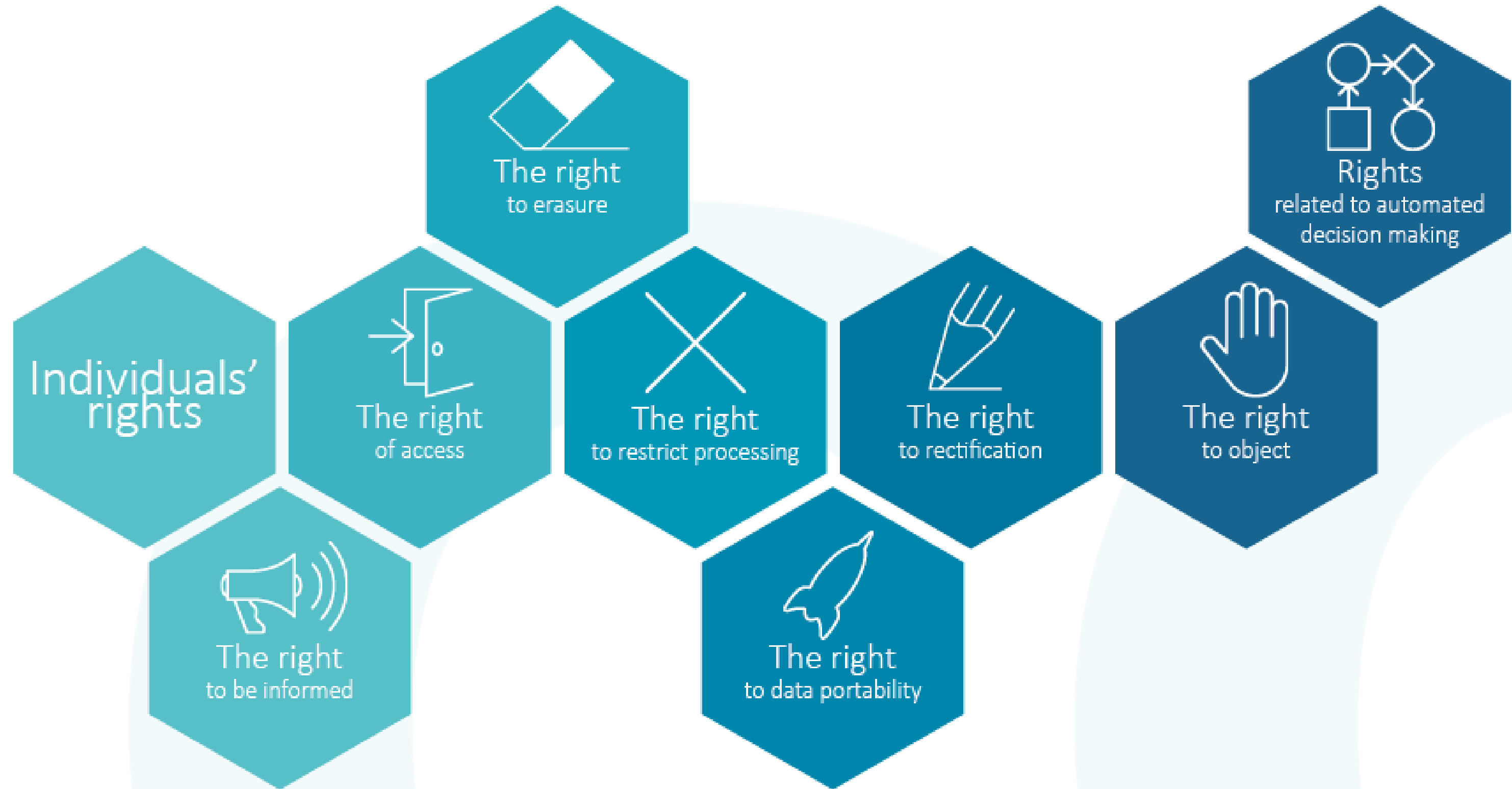
Integrity and confidentiality: Personal data must be processed securely



Accountability: You shall be able to demonstrate compliance with the above



Information rights



The Information Commissioner's Office (ICO)

What do we do?

Provide guidance on and enforce data protection regulation

- UK General Data Protection Regulation (UK GDPR)
- Data Protection Act 2018 (DPA 2018)
- Data (Use and Access) Act 2025 (DUAA)



Consult government during legislative reform

Work with companies via Sandbox and ad-hoc engagements

Investigate DP violations and issues warnings, reprimands, and fines



Not your average watchdog

What do we do?

Outsourcing firm Capita fined £14m after millions had data stolen



REUTERS

Imran Rahman-Jones

Technology reporter

15 October 2025

The UK's data watchdog has fined outsourcing firm Capita £14m after the personal data of 6.6 million people was stolen in a cyber-attack.

TikTok fined £12.7m for misusing children's data

4 April 2023



NURPHOTO

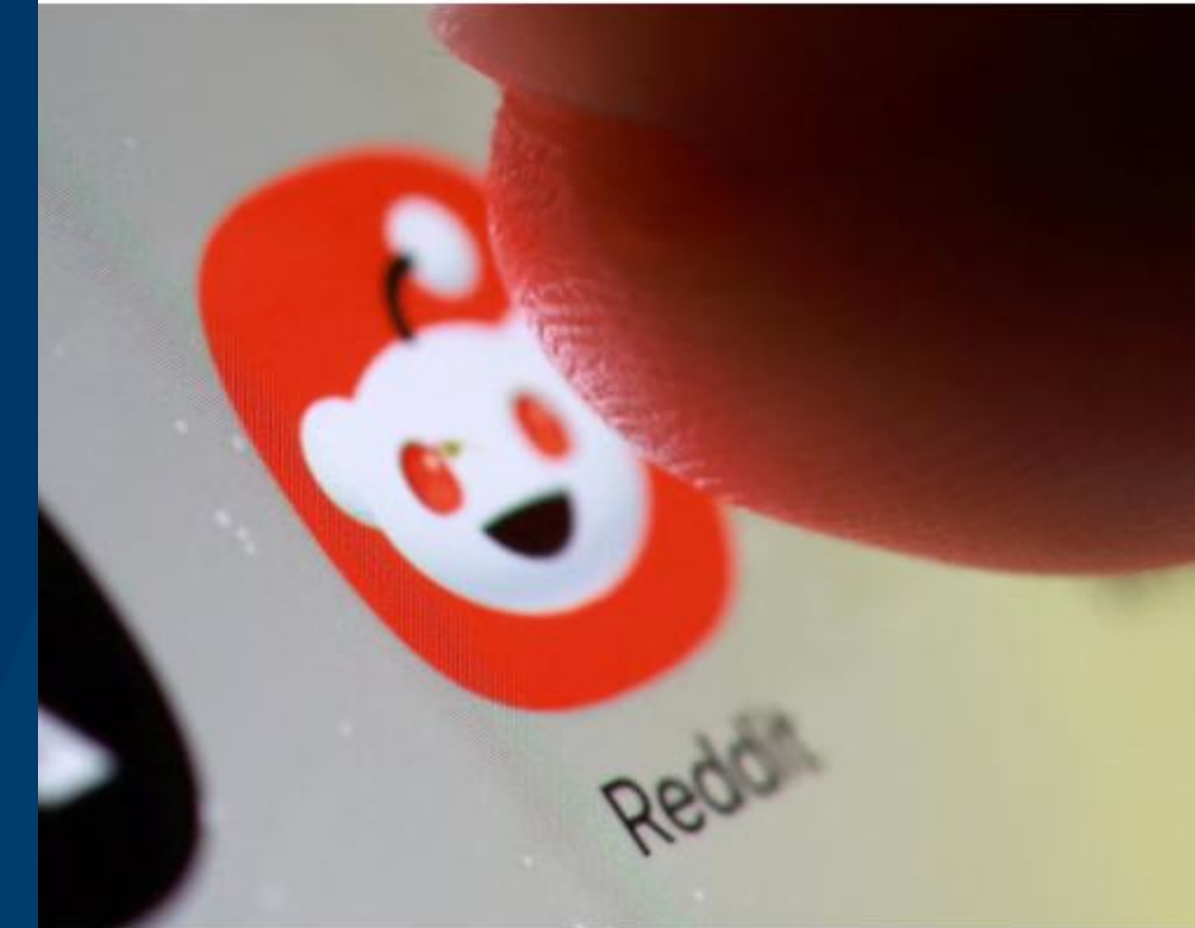
Shiona McCallum, Tom Gerken & Zoe Kleinman

BBC News

TikTok has been fined £12.7m by the UK's data watchdog for failing to protect the privacy of children.

Reddit fined £14.5m in UK over use of under-13s' data

Information Commissioner's Office imposes largest fine yet for a breach of children's privacy



Reddit said it did not require users to share information about their identities 'because we are deeply committed to their privacy and safety'. Photograph: Dado Ruvic/Reuters

The UK information regulator has fined the social news service [Reddit](#) £14.5m for using the data of children under the age of 13 unlawfully and potentially exposing them to inappropriate and harmful content.

ICO and AI

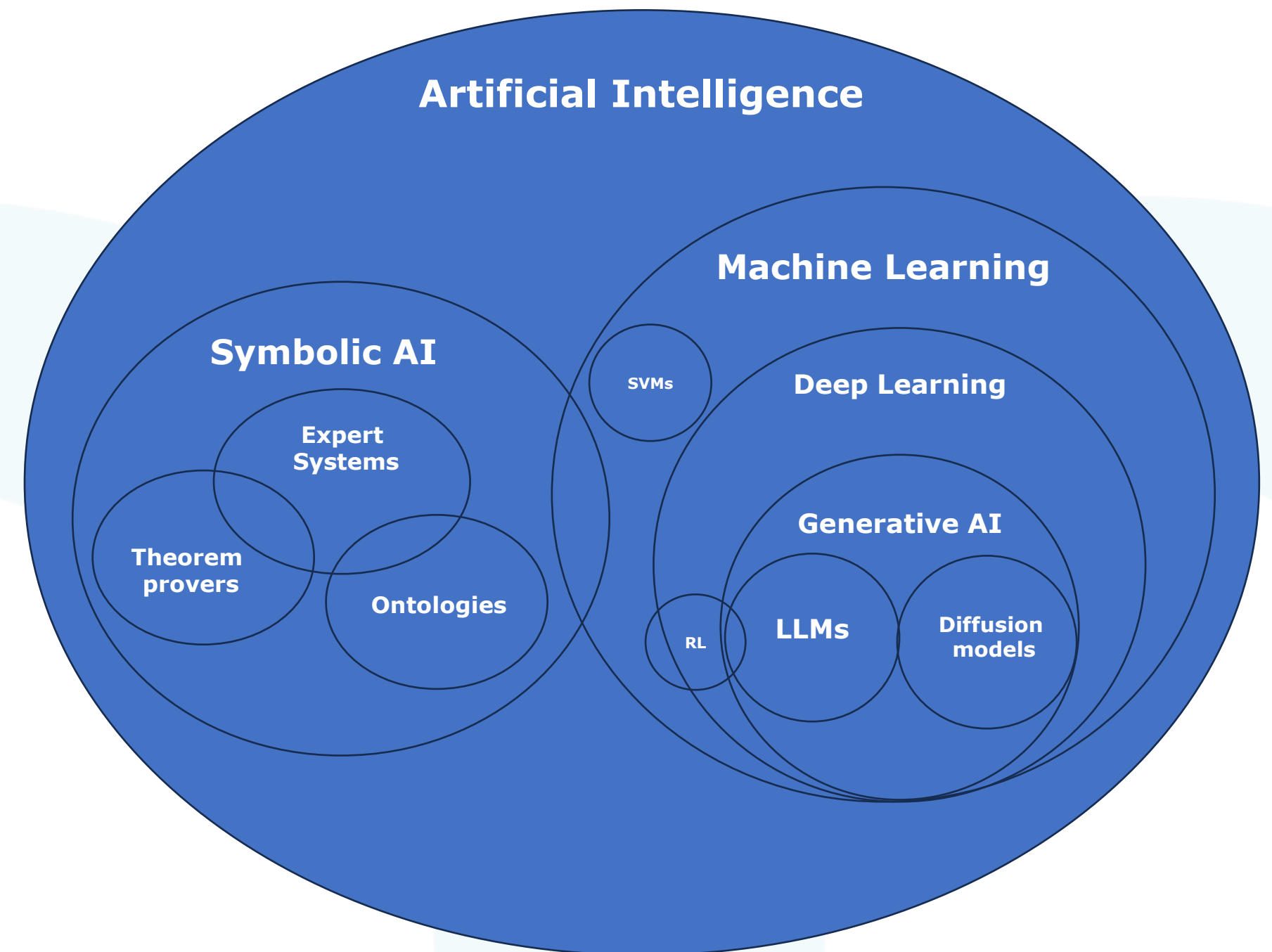


So, what about AI?

What is AI?

Approach view (how are we solving it?)

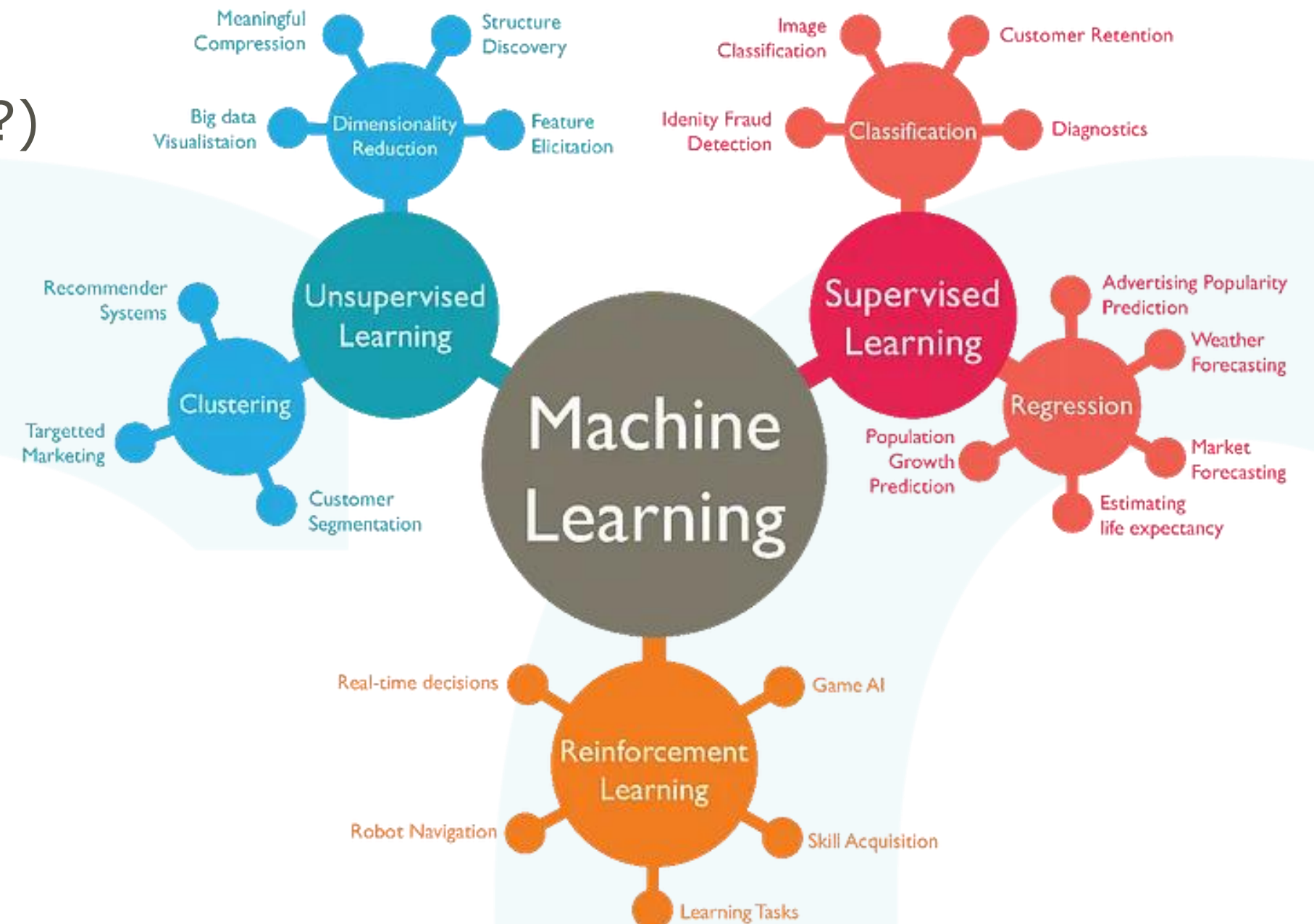
Task view (what problem are we solving?)



What is AI?

Approach view (how are we solving it?)

Task view (what problem are we solving?)



Large Language Models

Basic idea: predict the next word based on the previous ones (context)

Context is key - with enough information, the next word can be predicted with high accuracy.

(Small) Language Modeling

Context: “advise”	Next word probability
you	0.0235
the	0.0157
on	0.0038
...	...
...	...
zebra	0.0001

(Small) Language Modeling

Context: “advise”	Next word probability	Context: “they advise”	Next word probability
you	0.0235	on	0.0531
the	0.0157	the	0.0451
on	0.0038	you	0.0036
...	...	...	...
...	...	...	...
zebra	0.0001	zebra	0.0000

Large Language Models

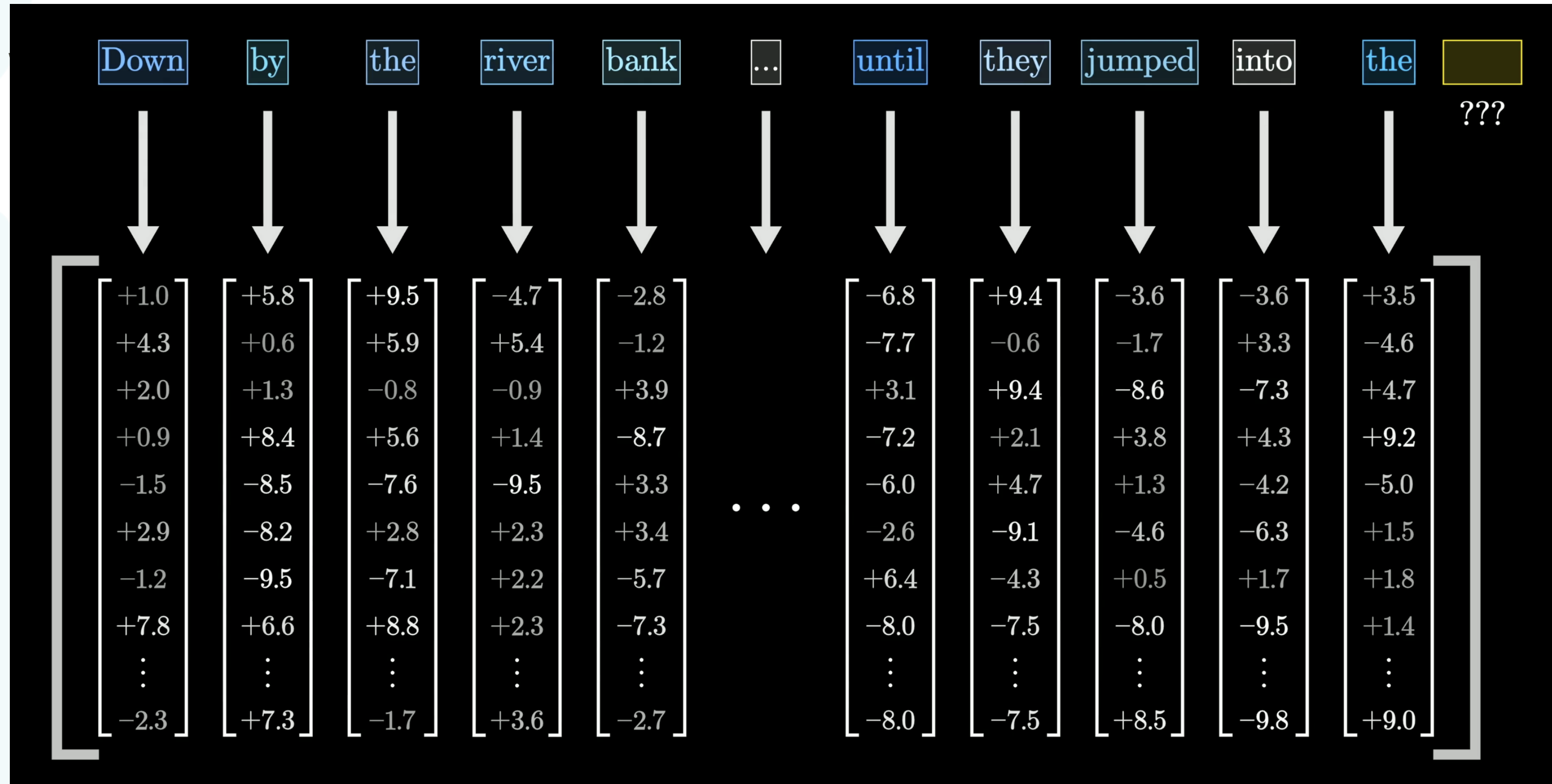
What changed?

Use “dense” representations (embeddings)

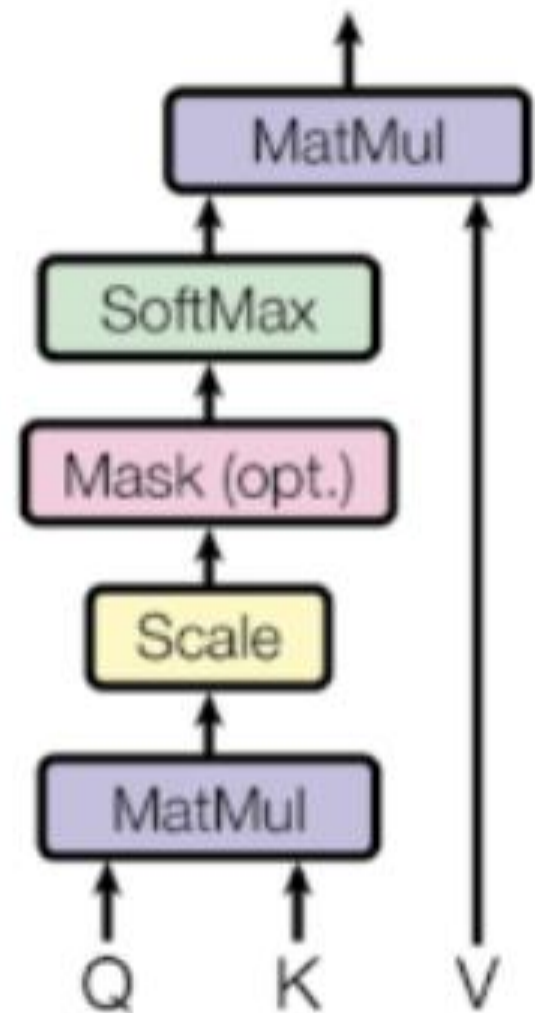
(Very) Deep Neural Networks to solve the optimisation

Use “attention mechanism” to capture word-word interactions

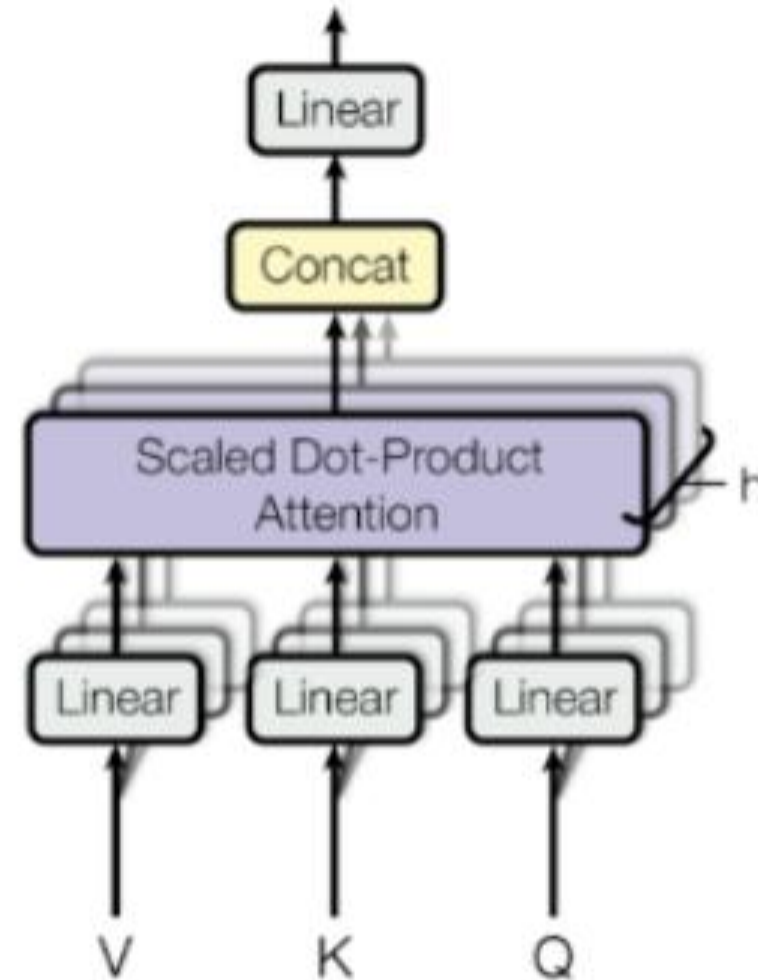
Large Language Models



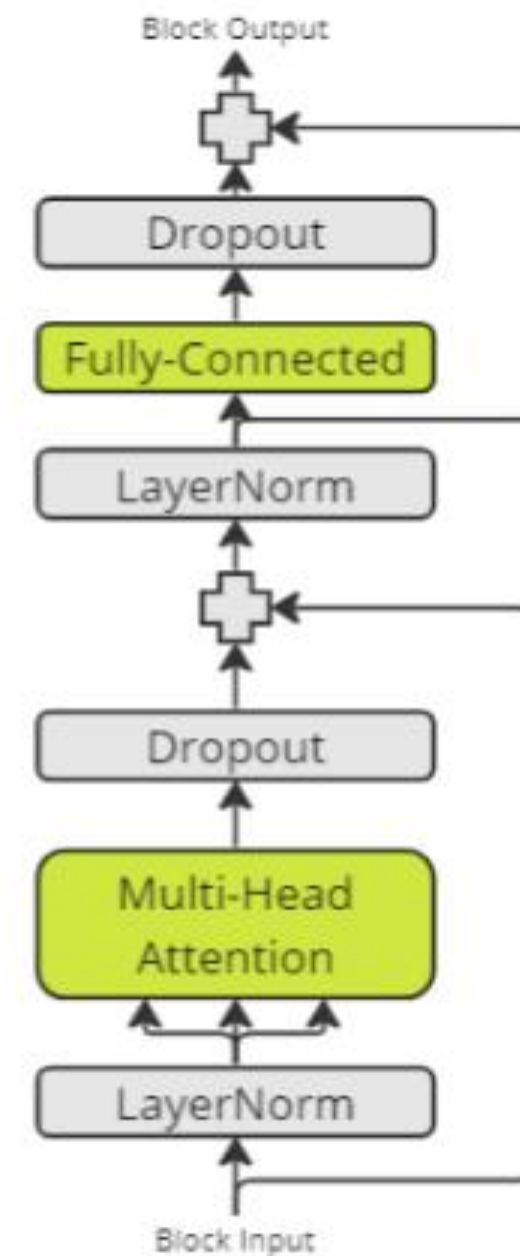
Large Language Models



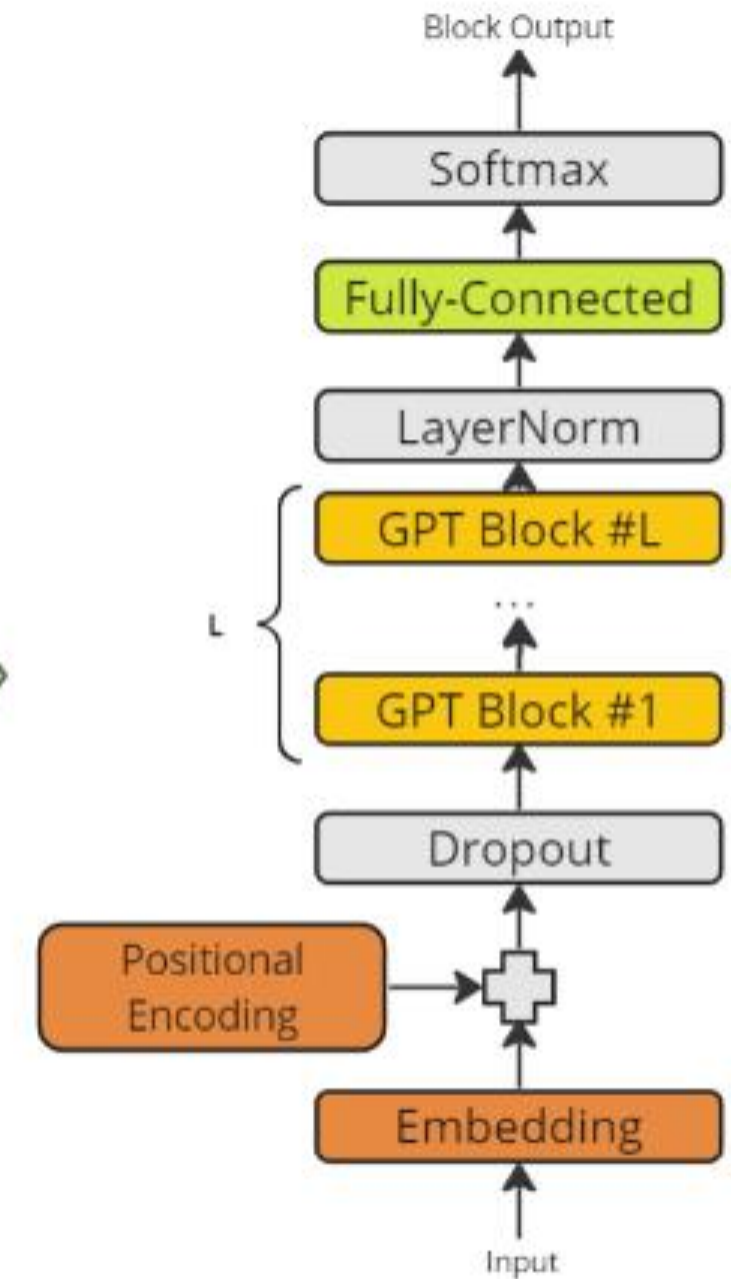
Scale Causal Attention



Multi-Head Attention



Transformer Block



GPT Model

Large Language Models

Getting to ChatGPT

Supervised Fine-tuning (SFT) / Instruction Tuning

Reinforcement Learning from Human Feedback (RLHF)

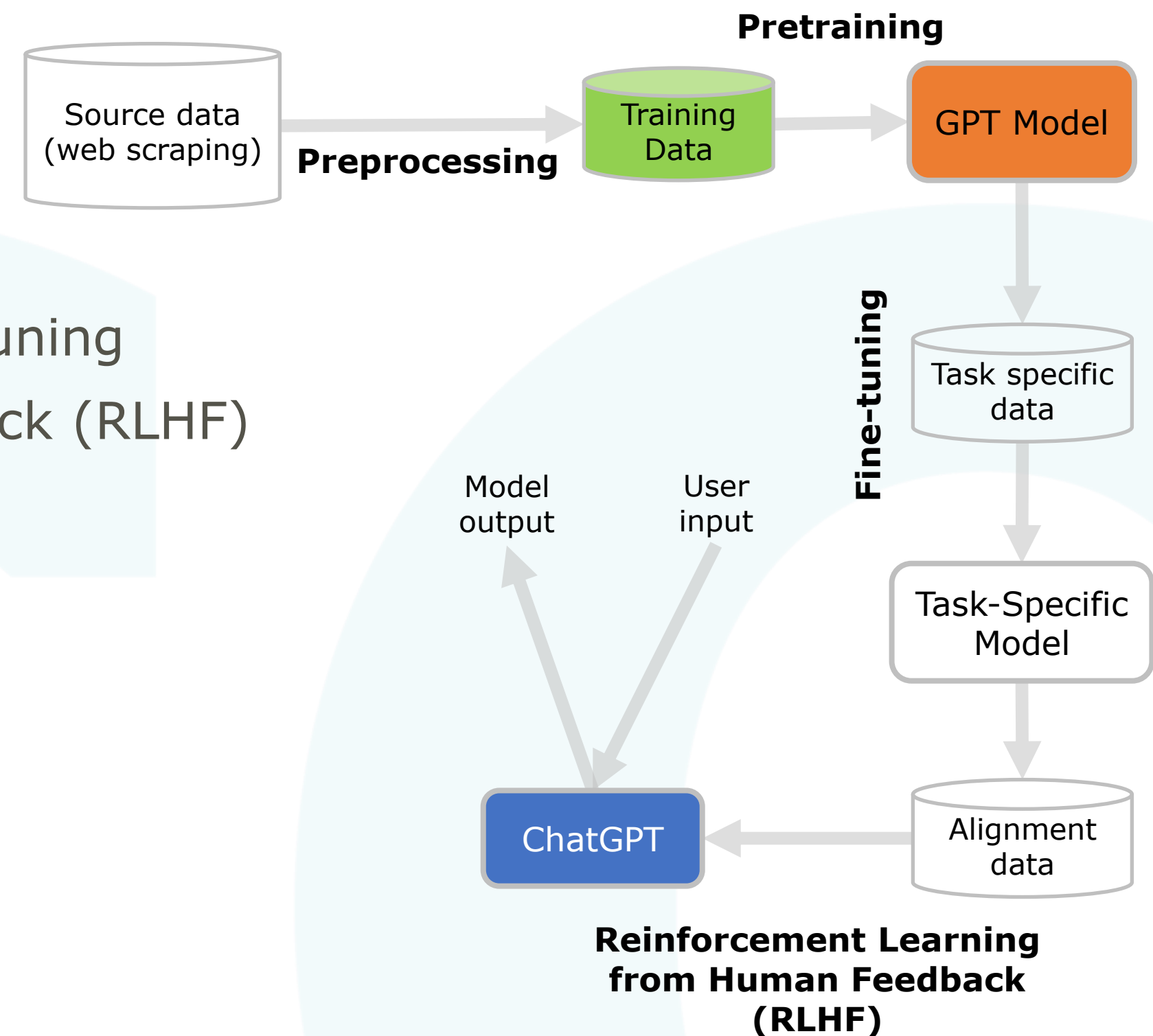
Scale:

Train on Trillions of words (tokens)

Using Billions of parameters

For months

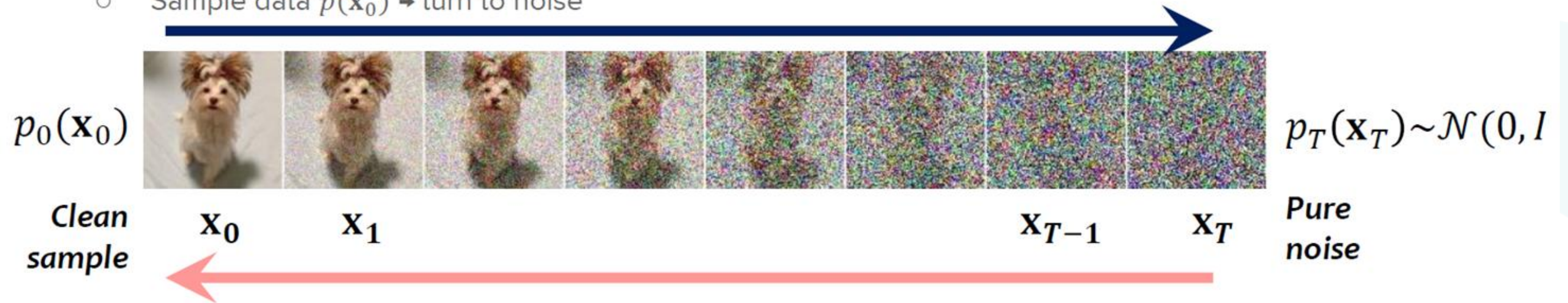
Costing 10s-100s \$M



Diffusion Models

- **Forward / noising process**

- Sample data $p(\mathbf{x}_0) \rightarrow$ turn to noise



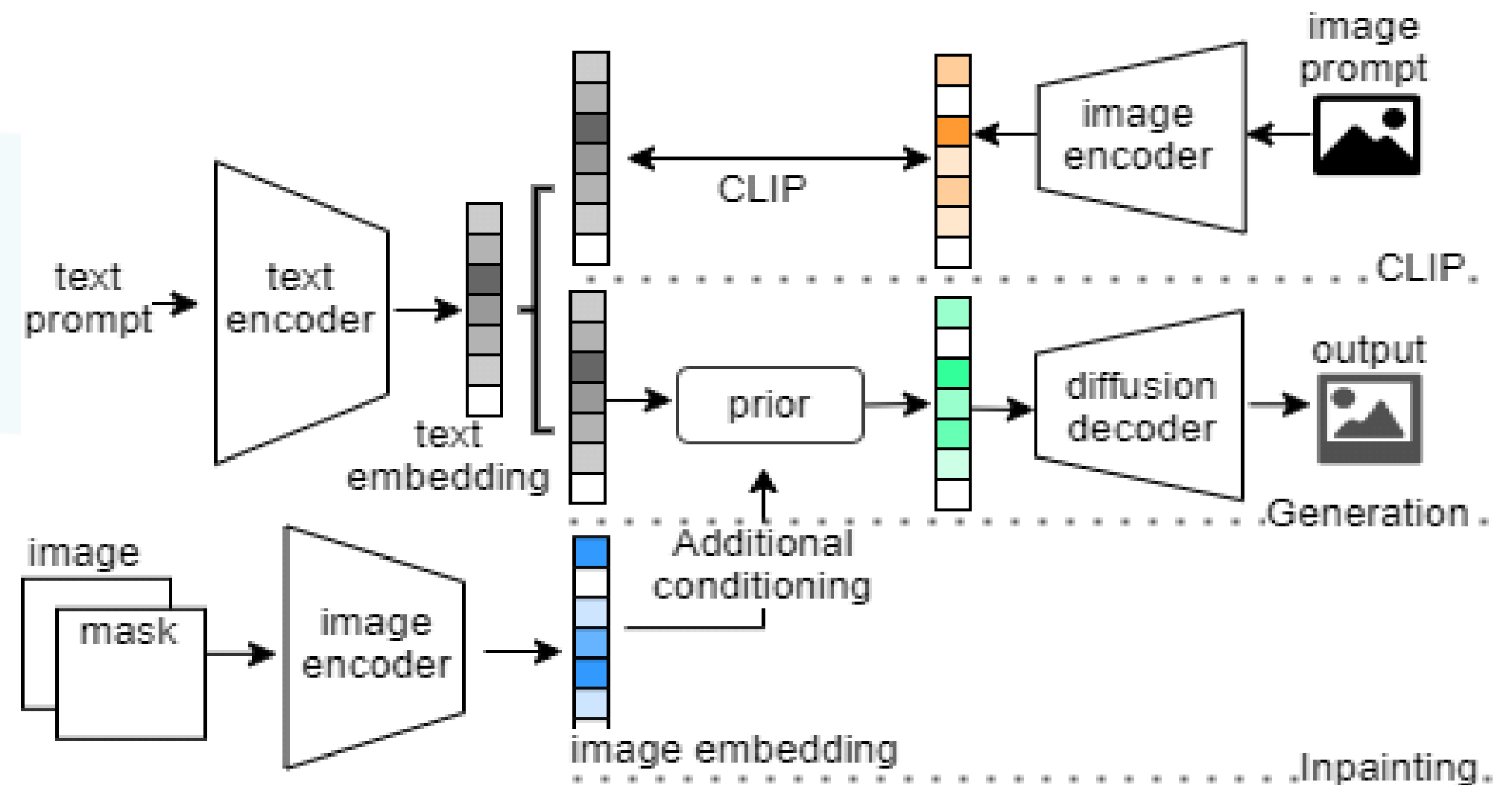
- **Reverse / denoising process**

- Sample noise $p_T(\mathbf{x}_T) \rightarrow$ turn into data

Diffusion Models

Train on image captions

Can be “conditioned” on
Text (image generation)
Images (image editing)

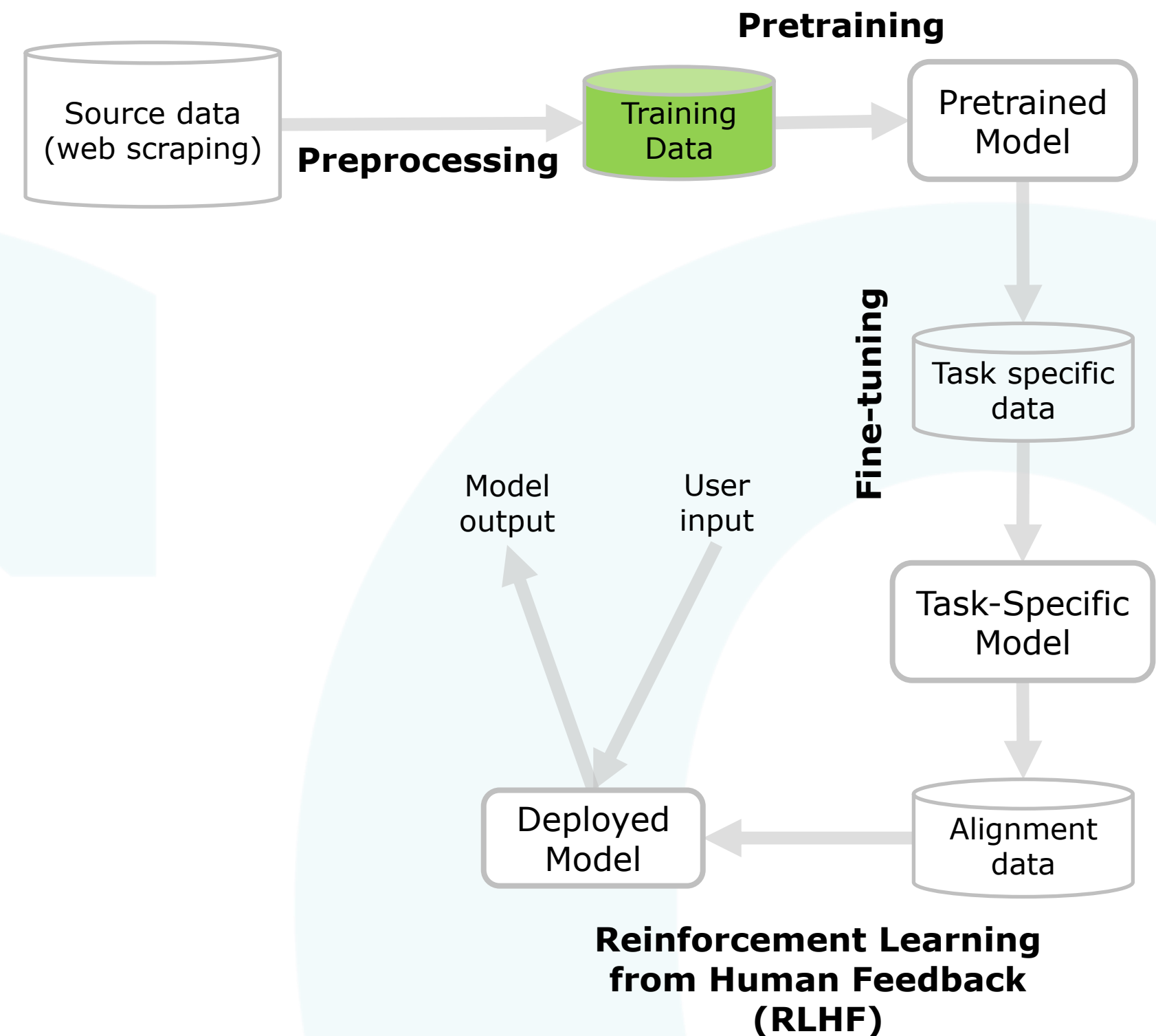


LLM Lifecycle & Data Protection

Personal data appears throughout the lifecycle

Processing activities:

- Web scraping
- Preprocessing
- Storing
- Training (pretraining, fine-tuning, alignment)
- Evaluation
- Inference
- Session storage
- Memory (personalisation)
- Tool use



Our 2025/2026 AI and biometrics strategy

- **Set clear expectations for responsible AI** through a statutory code of practice for organisations developing or deploying AI and automated decision-making
- Secure public confidence in **generative AI foundation models** by working with developers to ensure they use people's information responsibly and lawfully in training these models;
- Ensure that automated decision-making (ADM) systems are governed and used in a way that is fair to people, focusing on **ADM use in recruitment and in public services**; and
- Ensure the fair and proportionate **use of facial recognition technology (FRT) by the police**, working with law enforcement to ensure that the technology is effective and people's rights are protected.
- Anticipate and act on **emerging AI risks**, including agentic AI that raises questions around accountability and redress, and AI systems that make speculative inferences about people's intentions or emotions based on their physical or behavioural characteristics.

Questions

Group discussions

WeHire has built HAIRe, an AI agent that help employers with their hiring process.

HAIRe can scan applicant CVs, prioritise candidates for interviews, schedule the interviews, conduct the first screening, score them by analysing the (video) responses.

WeHire claims their software has been extensively tested and is shown to outperform human interviewers and hiring managers.

The Information Commissioner has asked you to engage with WeHire to understand whether there are any significant DP concerns.

I will play the DPO and CTO of WeHire and answer questions from each group.